Batch Arguments with Optimal Communication

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$\mathrm{Prove}\left(\mathrm{crs}, M, 1^t, \left(w_1, \ldots, w_k\right)\right) \to \pi$

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Somewhere extraction for every polynomial-size adversary $\mathcal{A}$ and index i:

$$\Pr\left[\begin{array}{c}\mathrm{Verify}(\mathrm{crs}, M, t, k, \pi) = 1 \\ M(i, w) \text{ rejects}\end{array} : \begin{array}{c}(\mathrm{crs}, \tau) \leftarrow \mathrm{Gen}(1^\lambda, i) \\ (M, 1^t, k, \pi) \leftarrow \mathcal{A}(\mathrm{crs}) \\ w \leftarrow \mathrm{Extract}(\tau, \pi)\end{array}\right] \leq \mathrm{negl}(\lambda)$$

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this work	$m + \mathrm{poly}(\lambda)$	$\mathrm{poly}(\lambda)$	unbounded depth: the proof grows only linearly with the number of levels

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SNARGs for P

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Assuming LWE (polynomial modulus-to-noise ratio), there is a fully local SEH with hash value size $m + \mathrm{poly}(\lambda)$ and key size $\mathrm{poly}(\lambda)$.

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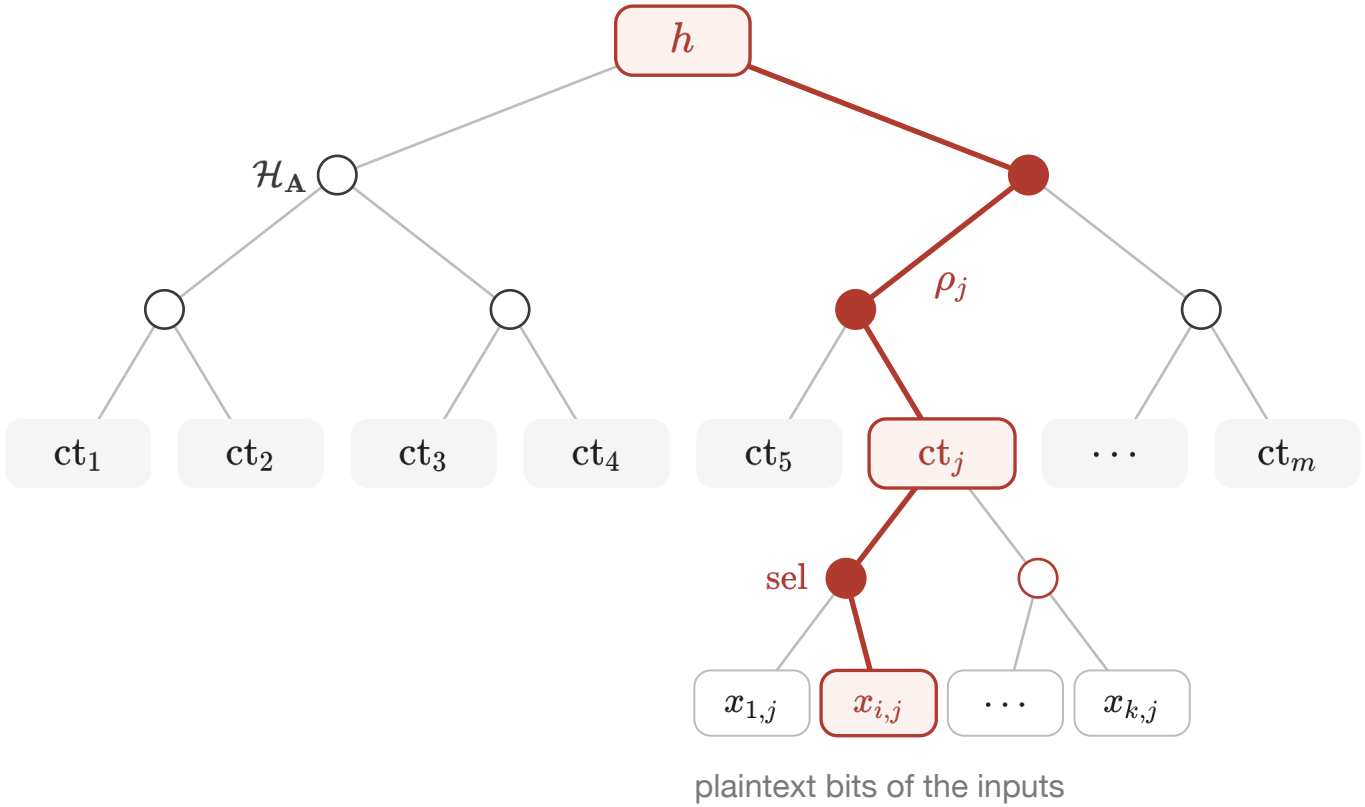
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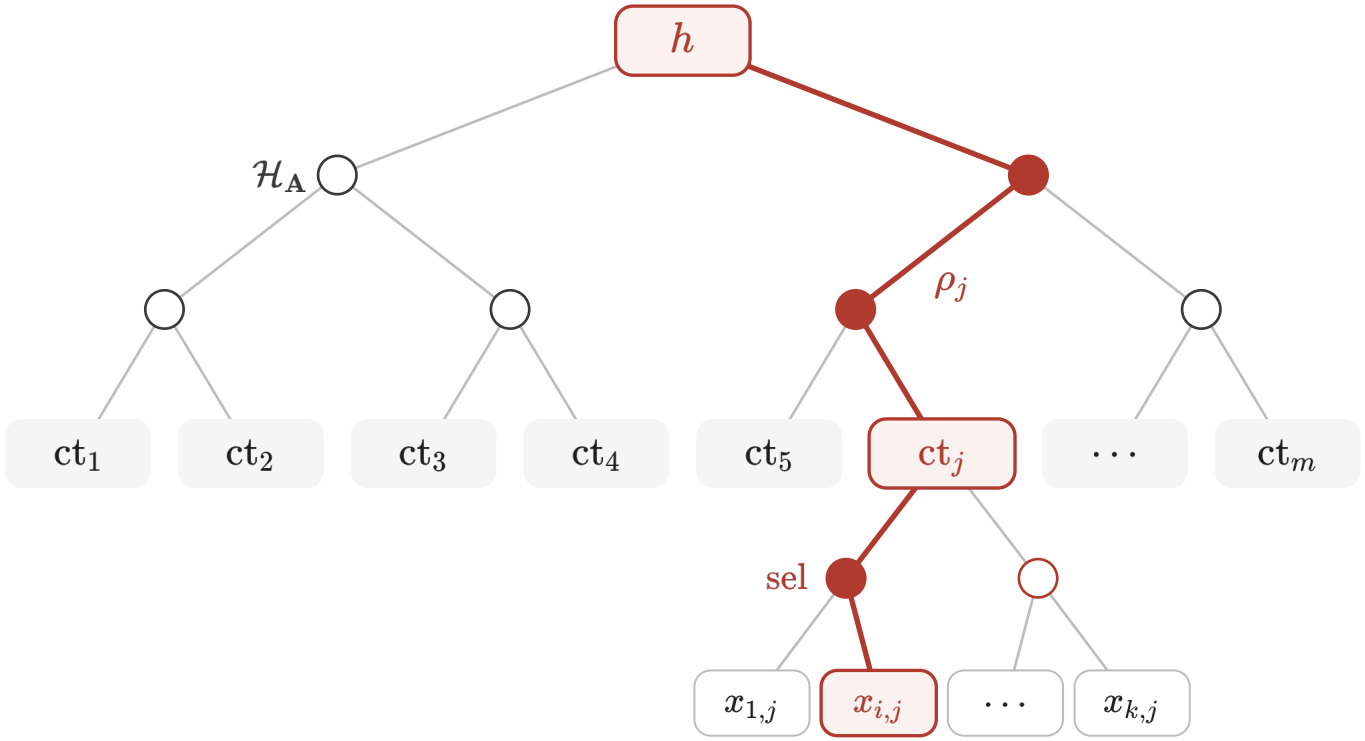
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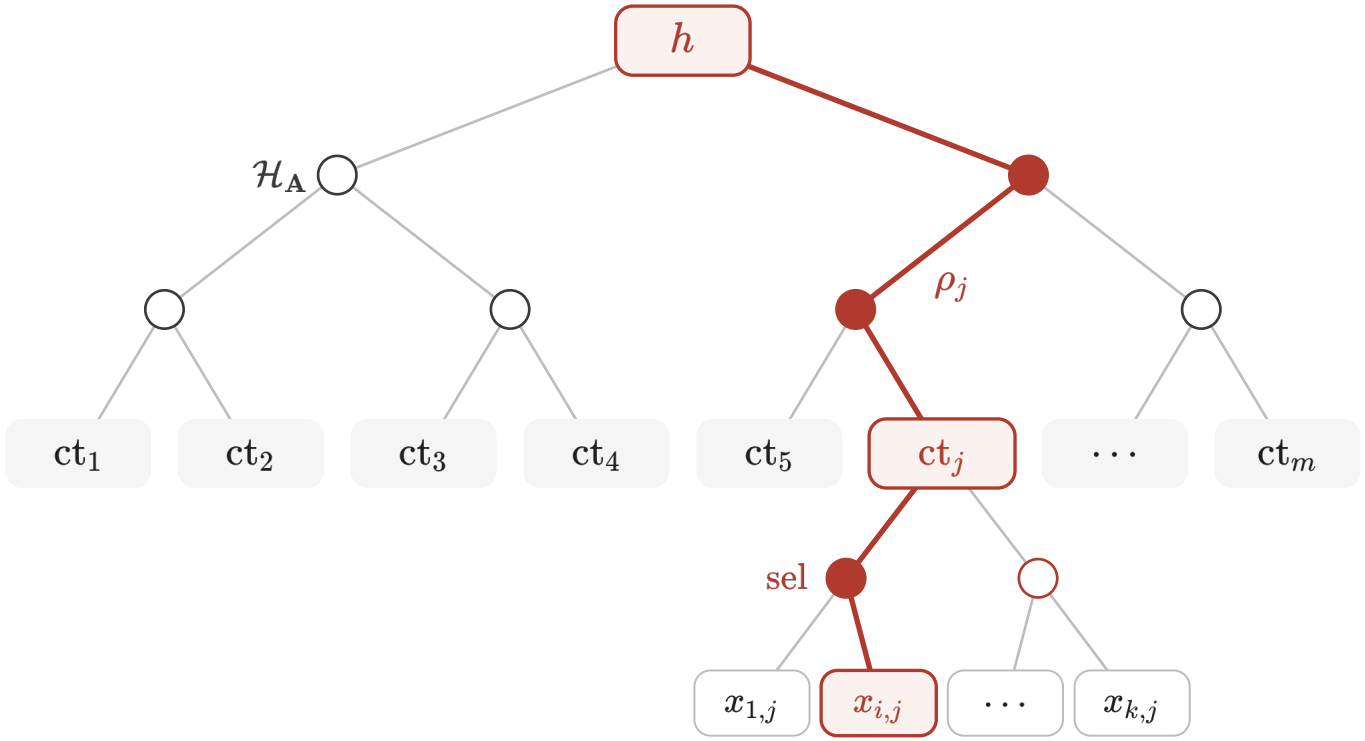
plaintext bits of the inputs

Selection node homomorphic, [HW15]

$$y \leftarrow \mathrm{Eval}\big(\mathrm{sel},\ \mathrm{Enc}(\mathrm{pk}, i^*_\ell),\ y_0, y_1\big)$$

encrypts $y_{i^*_\ell}$, the child picked by bit ℓ of i^*. Leaves are plaintext bits, so ct_j encrypts $x_{i^*,j}$; the key hides i^*.

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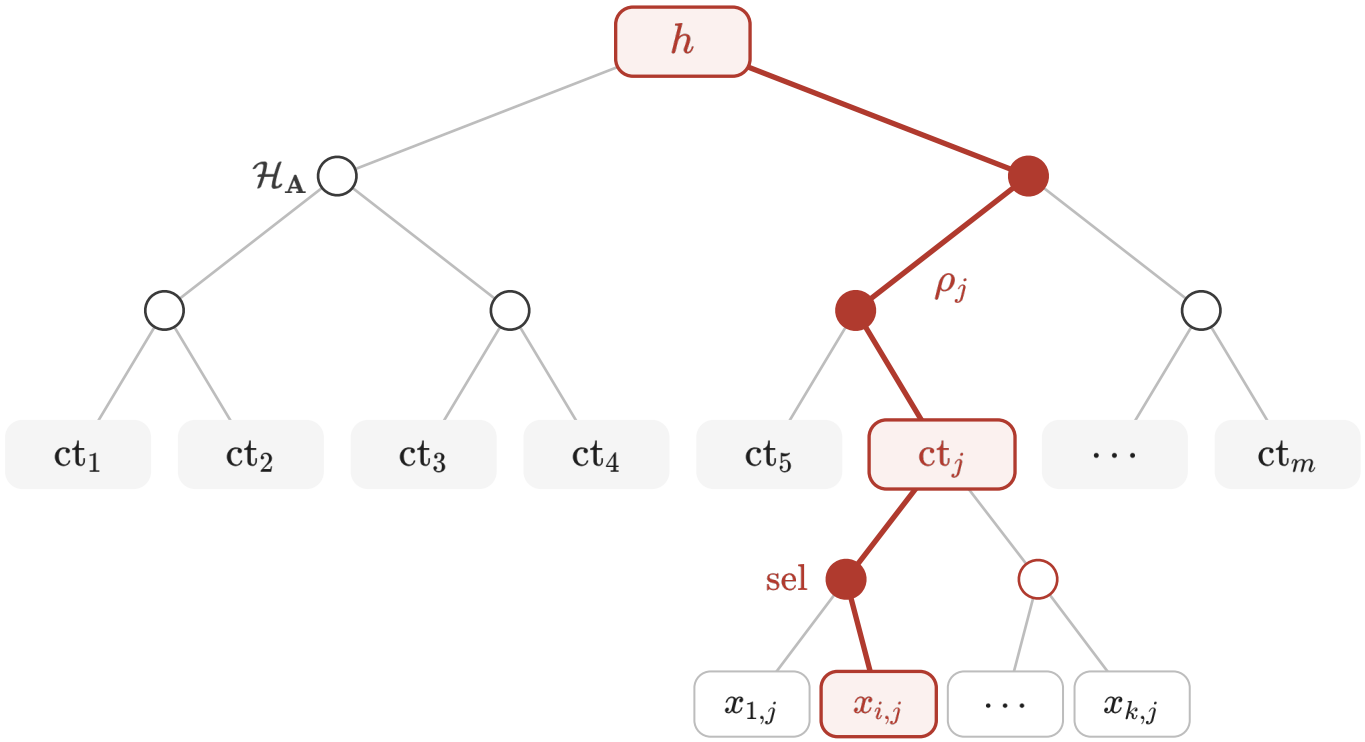
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$e_j \leftarrow \mathrm{Compress}(\mathrm{cpk}, j, \mathrm{ct}_j, \rho_j)$,
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$e_j \oplus d_j = x_{i^*,j}$ (next slide); openings are checked against h in time $\mathrm{poly}(\lambda)$.

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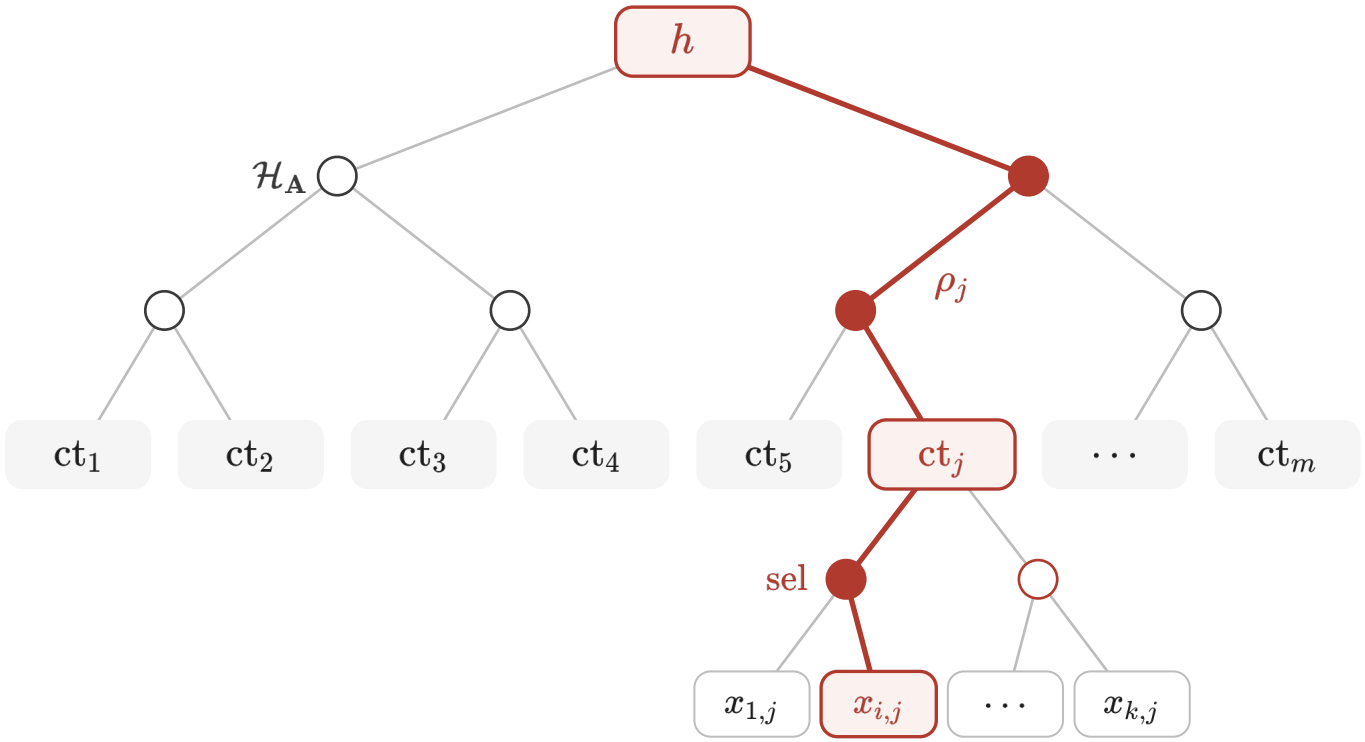
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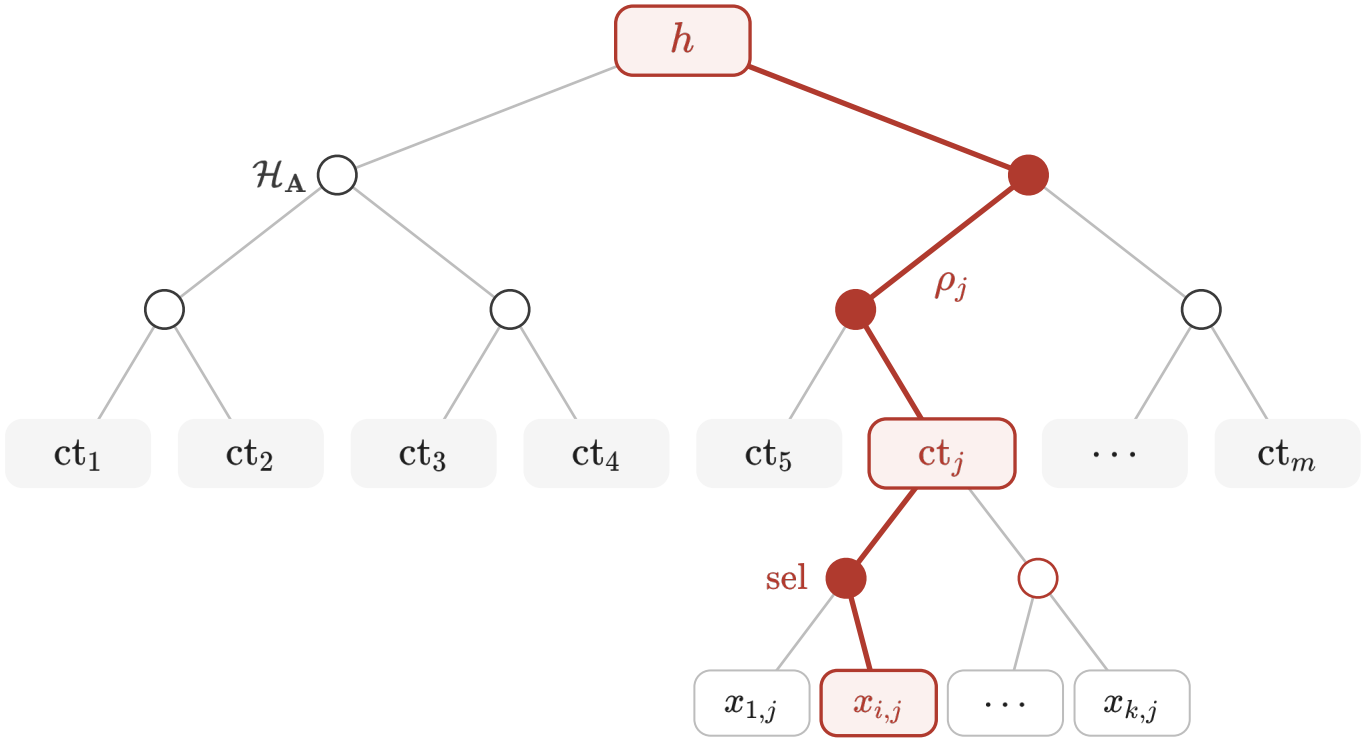
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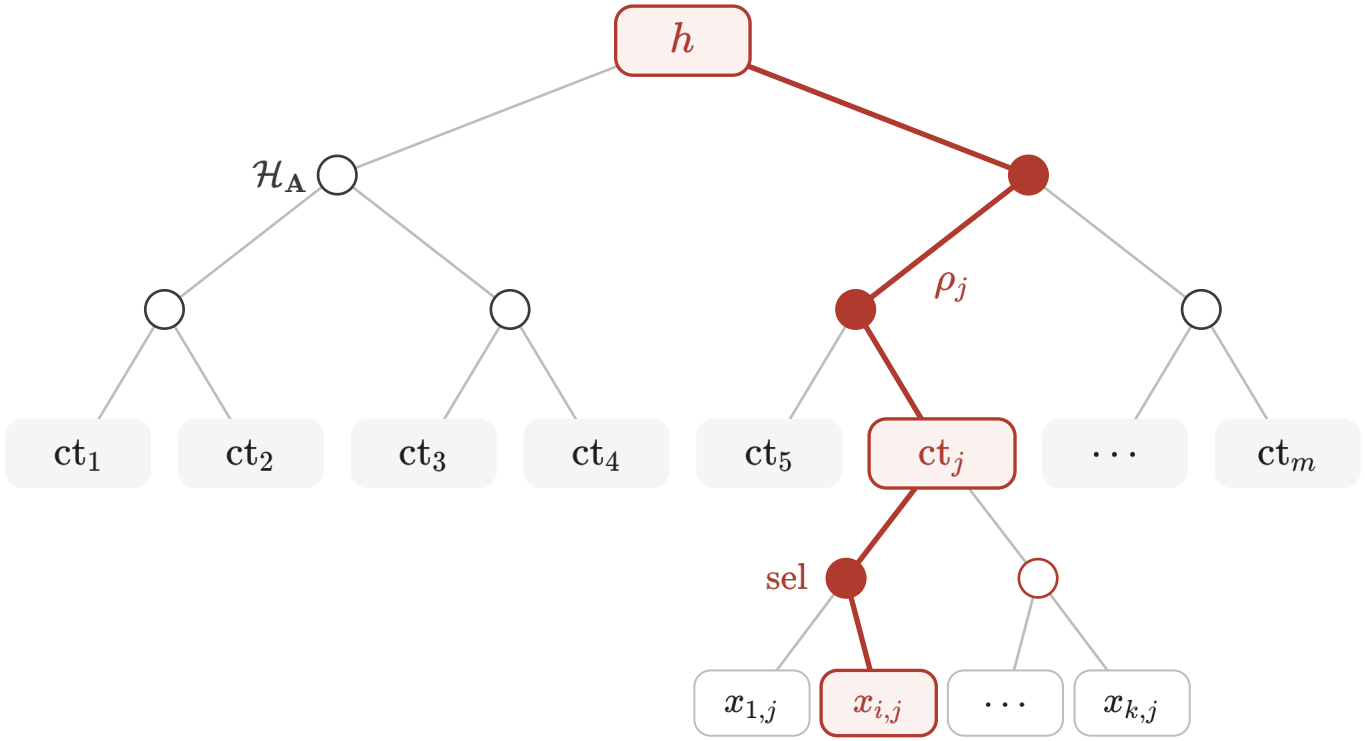
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trapdoor: the FHE secret key

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Decode$\left(i, \mathbf{h}, \mathbf{S}\right)$: $d_i := \mathrm{MSB}\left(\mathbf{u}_1^t\mathbf{B}_i\mathbf{S}\mathbf{h}\right)$ (computable from $\mathbf{S}$, $\mathbf{h}$ and i alone)

2-to-1 ciphertext compression [AMR25]

Almost-linear decryption [GSW] secret key $\mathbf{t} \in \mathbb{Z}_q^n$; a ciphertext $\mathbf{c} \in \mathbb{Z}_q^n$ satisfies $\mathrm{MSB}(\mathbf{t}^t\mathbf{c}) = \mathsf{msg}$

Compression Keys $\mathbf{A} \leftarrow \mathbb{Z}_q^{n\times 2N}$, $\mathbf{B}_0, \mathbf{B}_1 \leftarrow \{0,1\}^{N\times N}$, $\mathbf{S} \leftarrow \mathbb{Z}_q^{N\times n}$, $\mathbf{T} := \begin{pmatrix} \mathbf{t}^t \\ 0 \end{pmatrix} \in \mathbb{Z}_q^{N\times n}$, Gaussian $\mathbf{E}_0, \mathbf{E}_1$:

$$\mathbf{D}_0 := \mathbf{B}_0\mathbf{SA} - (\mathbf{TG}, 0) + \mathbf{E}_0 \qquad \mathbf{D}_1 := \mathbf{B}_1\mathbf{SA} - (0, \mathbf{TG}) + \mathbf{E}_1$$

(by LWE, $(\mathbf{D}_0, \mathbf{D}_1) \approx_c$ uniform: the public key hides $\mathbf{t}$)

Hash$(\mathbf{c}_0, \mathbf{c}_1)$: $\mathbf{h} := \mathscr{H}_{\mathbf{A}}(\mathbf{c}_0, \mathbf{c}_1) = \mathbf{Ag} \in \mathbb{Z}_q^n$, where $\mathbf{g} := \begin{pmatrix} \mathbf{G}^{-1}(\mathbf{c}_0) \\ \mathbf{G}^{-1}(\mathbf{c}_1) \end{pmatrix} \in \{0,1\}^{2N}$ (the SIS hash)

Compress$(i, \mathbf{c}_0, \mathbf{c}_1)$: $\mathbf{f}_i := \mathbf{D}_i\mathbf{g} \approx \mathbf{B}_i\mathbf{SAg} - \mathbf{TGG}^{-1}(\mathbf{c}_i) = \mathbf{B}_i\mathbf{Sh} - \mathbf{Tc}_i$, output $e_i := \mathrm{MSB}(\mathbf{u}_1^t\mathbf{f}_i)$ ($\mathbf{u}_1^t\mathbf{x}$: first coordinate of $\mathbf{x}$; $i \in \{0,1\}$)

Decode$(i, \mathbf{h}, \mathbf{S})$: $d_i := \mathrm{MSB}(\mathbf{u}_1^t\mathbf{B}_i\mathbf{Sh})$ (computable from $\mathbf{S}$, $\mathbf{h}$ and i alone)

Correctness

$$e_i \oplus d_i \approx \mathrm{MSB}(\mathbf{u}_1^t\mathbf{B}_i\mathbf{Sh} - \mathbf{t}^t\mathbf{c}_i) \oplus \mathrm{MSB}(\mathbf{u}_1^t\mathbf{B}_i\mathbf{Sh}) \approx \mathrm{MSB}(\mathbf{u}_1^t\mathbf{B}_i\mathbf{Sh}) \oplus \mathrm{MSB}(\mathbf{t}^t\mathbf{c}_i) \oplus \mathrm{MSB}(\mathbf{u}_1^t\mathbf{B}_i\mathbf{Sh}) = \mathrm{MSB}(\mathbf{t}^t\mathbf{c}_i) = \mathsf{msg}_i$$

Open problem:
Optimal BARGs without LWE?

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Nicolò Cesa-Bianchi
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Greg Gluch
Massachusetts Institute of Technology

Sam Gunn
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Andrew Ilyas
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Open problem:
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Thank you!

Questions?

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